



PARALLEL PROGRAMMING

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<https://github.com/canbula/ParallelProgramming/>



qi9tndw

Parallel Programming

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Course Overview

Parallel Programming (Teams Code: qi9tndw)

We are going to learn the basics of asynchronous programming, creating multiple threads and processes in this course. Python is preferred as the programming language for the applications of this course.

Required Text

Python Concurrency with asyncio, Manning, *Matthew Fowler*

A Practical Approach to High-Performance Computing, Springer, *Sergei Kurgalin – Sergei Borzunov*

Python Parallel Programming Cookbook, Packt, *Giancarlo Zaccone*

Course Materials

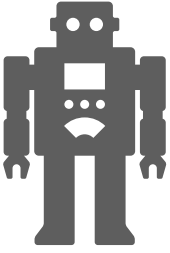
- Python 3.x (Anaconda is preferred)
- Jupyter Notebook from Anaconda
- Pycharm from JetBrains / Microsoft Visual Studio Code
- PC with a Linux distro or a Linux terminal in Windows 10/11.

Course Schedule

Week	Subject	Week	Subject
01	Data Structures in Python	08	Deadlock and Semaphore
02	Functions and Decorators in Python	09	Barriers and Conditions
03	Coroutines and Concurrency with asyncio	10	Creating Processes with multiprocessing
04	IO-bound Problems and Concurrency	11	Pipes and Queues
05	Creating Threads in Python with threading	12	CPU-bound Problems and Parallelism
06	Global Interpreter Lock and JIT Compiler	13	Creating Clusters
07	Protecting Resources with Lock	14	Load Balancing with Containers

Natural Languages vs. Programming Languages

A language is a tool for expressing and recording thoughts.



Computers have their own language called **machine** language. Machine languages are created by humans, no computer is currently capable of creating a new language. A complete set of known commands is called an instruction list (IL).

The difference is that human languages developed naturally. They are still evolving, new words are created every day as old words disappear. These languages are called **natural** languages.



Elements of a Language

- **Alphabet** is a set of symbols to build words of a certain language.
- **Lexis** is a set of words the language offers its users.
- **Syntax** is a set of rules used to determine if a certain string of words forms a valid sentence.
- **Semantics** is a set of rules determining if a certain phrase makes sense.



Machine Language vs. High-Level Language

The IL is the alphabet of a machine language. It's the computer's mother tongue.

High-level programming language enables humans to write their programs and computers to execute the programs. It is much more complex than those offered by ILs.

A program written in a high-level programming language is called a **source code**. Similarly, the file containing the source code is called the **source file**.

Compilation vs. Interpretation

There are two different ways of transforming a program from a high-level programming language into machine language:

Compilation: The source code is translated once by getting a file containing the machine code.

Interpretation: The source code is interpreted every time it is intended to be executed.

Compilation

- The execution of the translated code is usually faster.
- Only the user has to have the compiler. The end user may use the code without it.
- The translated code is stored using machine language. Your code are likely to remain your secret.



- The compilation itself may be a very time-consuming process
- You have to have as many compilers as hardware platforms you want your code to be run on.

Interpretation

- You can run the code as soon as you complete it, there are no additional phases of translation.
- The code is stored using programming language, not machine language. You don't compile your code for each different architecture.



- Don't expect interpretation to ramp up your code to high speed
- Both you and the end user have the interpreter to run your code.

What is Python?

Python is a widely-used, interpreted, object-oriented, and high-level programming language with dynamic semantics, used for general-purpose programming.

Python was created by Guido van Rossum. The name of the Python programming language comes from an old BBC television comedy sketch series called Monty Python's Flying Circus.



Guido van Rossum

Python Goals

- an **easy and intuitive** language just as powerful as those of the major competitors
- **open source**, so anyone can contribute to its development
- code that is as **understandable** as plain English
- **suitable for everyday tasks**, allowing for short development times



Why Python?



- easy to learn
- easy to teach
- easy to use
- easy to understand
- easy to obtain, install and deploy

Why not Python?



- low-level programming
- applications for mobile devices

Python Implementations

An implementation refers to a program or environment, which provides support for the execution of programs written in the Python language.

- **CPython** is the traditional implementation of Python and it's most often called just "Python".
- **Cython** is a solution which translate Python code into "C" to make it run much faster than pure Python.
- **Jython** is an implementation follows only Python 2, not Python 3, written in Java.
- **PyPy** represents a Python environment written in Python-like language named RPython (Restricted Python), which is actually a subset of Python.
- **MicroPython** is an implementation of Python 3 that is optimized to run on microcontrollers.

Start Coding with Python

- **Editor** will support you in writing the code. The Python 3 standard installation contains a very simple application named IDLE (Integrated Development and Learning Environment).
- **Console** is a terminal in which you can launch your code.
- **Debugger** is a tool, which launches your code step-by-step to allow you to inspect it.

first.py

```
print("Python is the best!")
```



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in-class quiz

Function Name

A function can cause some effect or evaluate a value, or both.

Where do functions come from?

- From Python itself
- From modules
- From your code

first.py

```
print("Python is the best!")
```

Argument

- Positional arguments
- Keyword arguments

```
print(*objects, sep=' ', end='\n', file=None, flush=False)
```

Print *objects* to the text stream *file*, separated by *sep* and followed by *end*. *sep*, *end*, *file*, and *flush*, if present, must be given as keyword arguments.

All non-keyword arguments are converted to strings like `str()` does and written to the stream, separated by *sep* and followed by *end*. Both *sep* and *end* must be strings; they can also be `None`, which means to use the default values. If no *objects* are given, `print()` will just write *end*.

The *file* argument must be an object with a `write(string)` method; if it is not present or `None`, `sys.stdout` will be used. Since printed arguments are converted to text strings, `print()` cannot be used with binary mode file objects. For these, use `file.write(...)` instead.

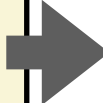
Output buffering is usually determined by *file*. However, if *flush* is true, the stream is forcibly flushed.

Literals

A literal is data whose values are determined by the literal itself. Literals are used to encode data and put them into code.

literals.py

```
print("7")  
print(7)  
print(7.0)  
print(7j)  
print(True)  
print(0b10)  
print(0o10)  
print(0x10)  
print(7.4e3)
```

- 
- String
 - Integer
 - Float
 - Complex
 - Boolean
 - Binary
 - Octal
 - Hexadecimal
 - Scientific Notation

Basic Operators

An operator is a symbol of the programming language, which is able to operate on the values.

Multiplication

```
print(2 * 3) Integer
print(2 * 3.0) Float
print(2.0 * 3) Float
print(2.0 * 3.0) Float
```

Division

```
print(6 / 3) Float
print(6 / 3.0) Float
print(6.0 / 3) Float
print(6.0 / 3.0) Float
```

Exponentiation

```
print(2**3) Integer
print(2**3.0) Float
print(2.0**3) Float
print(2.0**3.0) Float
```

Floor Division

```
print(6 // 3) Integer
print(6 // 3.0) Float
print(6.0 // 3) Float
print(6.0 // 3.0) Float
```

Modulo

```
print(6 % 3) Integer
print(6 % 3.0) Float
print(6.0 % 3) Float
print(6.0 % 3.0) Float
```

Addition

```
print(-8 + 4) Integer
print(-4.0 + 8) Float
```

Operator Priorities

An operator is a symbol of the programming language, which is able to operate on the values.

priorities.py

```
print(9 % 6 % 2)
print(2**2**3)
print(2 * 3 % 5)
print(-3 * 2)
print(-2 * 3)
print(-(2 * 3))
```

- + (unary)
- - (unary)
- ** (right-sided binding)
- *
- /
- //
- % (left-sided binding)
- + (binary)
- - (binary)

Variables

Variables are symbols for memory addresses.

Built-in Functions

The Python interpreter has a number of functions and types built into it that are always available. They are listed here in alphabetical order.

Built-in Functions			
A abs() aiter() all() anext() any() ascii()	E enumerate() eval() exec() F filter()	L len() list() locals() M map()	R range() repr() reversed() round() S

hex(x)

Convert an integer number to a lowercase hexadecimal string prefixed with “0x”. If x is not a Python `int` object, it has to define an `__index__()` method that returns an integer. Some examples:

```
>>> hex(255)
'0xff'
>>> hex(-42)
'-0x2a'
```

classmethod() compile() complex()	help() hex()	ord() P pow() print()	type() V vars()
D	I id()		

id(object)

Return the “identity” of an object. This is an integer which is guaranteed to be unique and constant for this object during its lifetime. Two objects with non-overlapping lifetimes may have the same `id()` value.

Identifier Names

For variables, functions, classes etc. we use identifier names. We must obey some rules and we should follow some naming conventions.

- Names are case sensitive.
- Names can be a combination of letters, digits, and underscore.
- Names can only start with a letter or underscore, can not start with a digit.
- Keywords can not be used as a name.



keyword — Testing for Python keywords

Source code: [Lib/keyword.py](#)

This module allows a Python program to determine if a string is a [keyword](#) or [soft keyword](#).

`keyword.iskeyword(s)`

Return `True` if `s` is a Python [keyword](#).

`keyword.kwlist`

Sequence containing all the [keywords](#) defined for the interpreter. If any keywords are defined to only be active when particular `__future__` statements are in effect, these will be included as well.

`keyword.issoftkeyword(s)`

Return `True` if `s` is a Python [soft keyword](#).

New in version 3.9.

`keyword.softkwlist`

Sequence containing all the [soft keywords](#) defined for the interpreter. If any soft keywords are defined to only be active when particular `__future__` statements are in effect, these will be included as well.

New in version 3.9.



Your First Homework



Week01/info_firstname_lastname.py

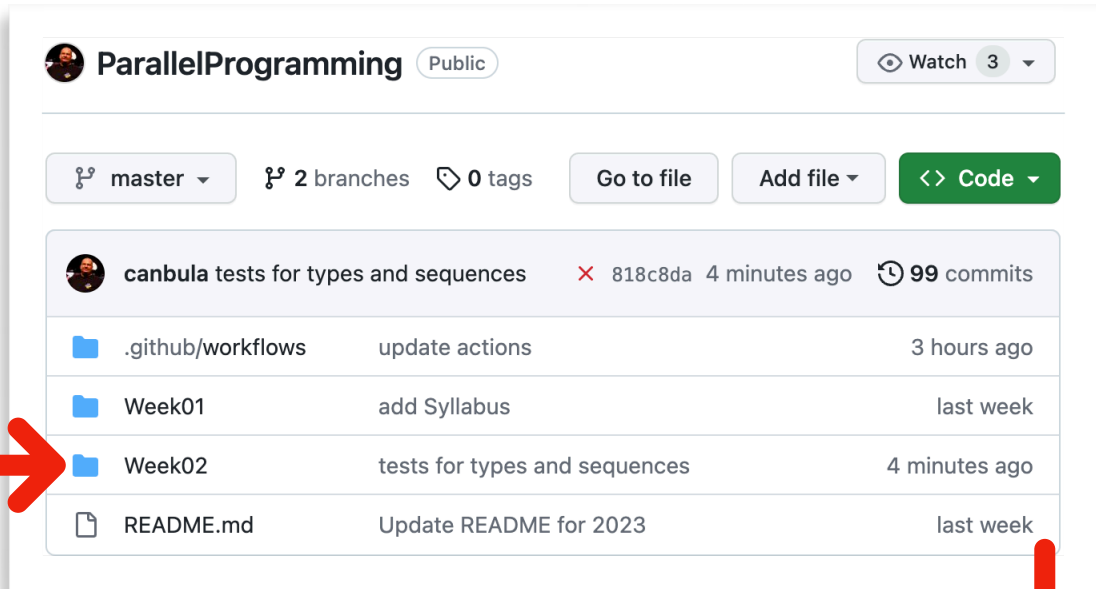
- ✓ A string variable with the name `student_id` that contains your student id.
- ✓ A string variable with the name `full_name` that contains your full name.



Week02/types_firstname_lastname.py

- ✓ An integer with the name: `my_int`
- ✓ A float with the name: `my_float`
- ✓ A boolean with the name: `my_bool`
- ✓ A complex with the name: `my_complex`

Your First Homework

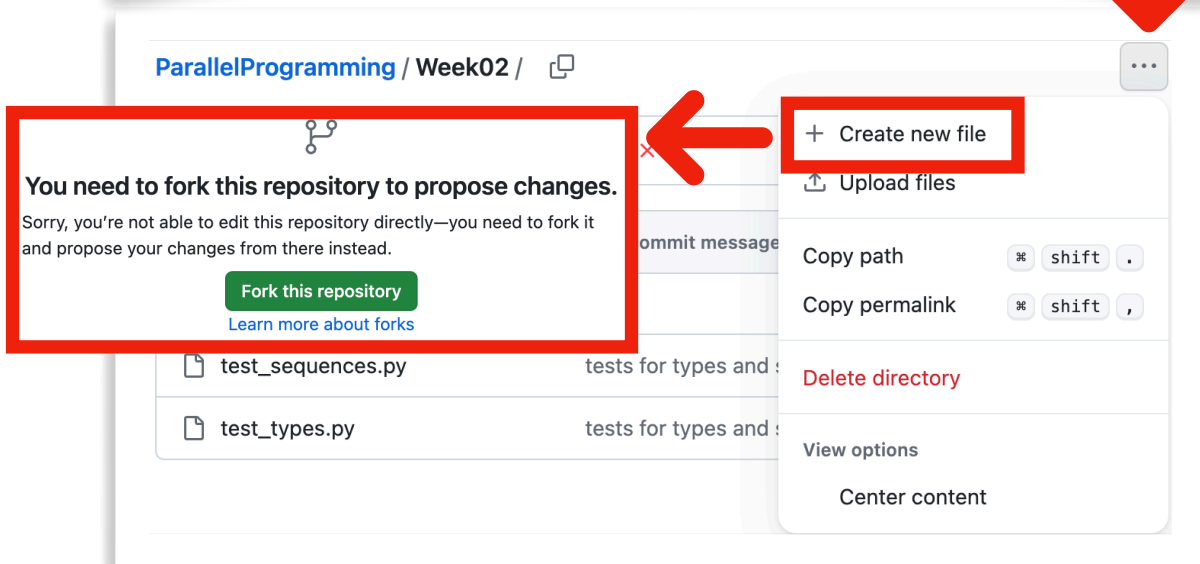


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canbula tests for types and sequences 818c8da 4 minutes ago 99 commits

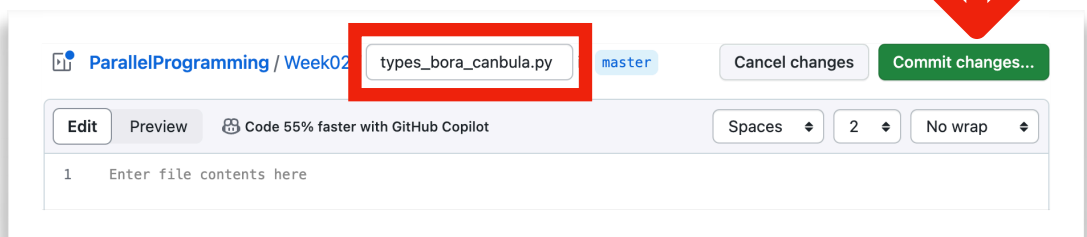
.github/workflows	update actions	3 hours ago
Week01	add Syllabus	last week
Week02	tests for types and sequences	4 minutes ago
README.md	Update README for 2023	last week



ParallelProgramming / Week02

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Sorry, you're not able to edit this repository directly—you need to fork it and propose your changes from there instead.
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ParallelProgramming / Week02 types_bora_canbula.py master Cancel changes Commit changes...

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1 Enter file contents here

Equality & Identity & Comparison

Equality

```
a = 4
print(a == 4) ✓
print(id(a) == id(4)) ✓
print(a is 4) ✓
print(id(a) is id(4)) ✗
```

The operators `is` and `is not` test for an object's identity: `x is y` is true if and only if `x` and `y` are the same object. An Object's identity is determined using the `id()` function. `x is not y` yields the inverse truth value. [4]

Left- or Right-sided?

```
x, y, z = 0, 1, 2
print(x == y == z) ✗
print(x == (y == z)) ✓
print((x == y) == z) ✗
```

Inequality

```
print(1 != 2) ✓
print(not 1 == 2) ✓
```

Comparison

```
print(1 < 2) ✓
print(1 <= 2) ✓
print(1 > 2) ✗
print(1 >= 2) ✗
```

Chaining

```
print(1 < 2 < 3) ✓
print(1 < 2 > 3) ✗
print(1 < 2 >= 3) ✗
print(1 < 2 <= 3) ✓
```

Updated Priority Table

Operator	Type
<code>+, -</code>	unary
<code>**</code>	binary
<code>*, /, //, %</code>	binary
<code>+, -</code>	binary
<code><, <=, >, >=</code>	binary
<code>!=, ==</code>	binary

QUESTION Using one of the comparison operators in Python, write a simple two-line program that takes the parameter `n` as input, which is an integer, prints **False** if `n` is less than 100, and **True** if `n` is greater than or equal to 100.

```
n = int(input())
print(n >= 100)
```

Conditional Execution

if statement

```
if n >= 100:
    print("The number is greater than or equal to 100.")
elif n < 0:
    print("The number is negative.")
else:
    print("The number is less than 100.")
```

Ternary Operator

```
msg = "The number is greater than or equal to 100." if n >= 100 else "The number is less than 100."
print(msg)
```

Loops

QUESTION

- The program generates a random number between 1 and 10.
- The user is asked to guess the number.
- The user is given feedback if the guess is too low or too high.
- The user is asked to guess again until the correct number is guessed.

```
r = random.randint(1, 10)
answer = False
while not answer:
    n = int(input("Enter a number: "))
    if n == r:
        print("You guessed it right!")
        answer = True
    elif n < r:
        print("Try a higher number.")
    else:
        print("Try a lower number.")
```

QUESTION

- The user is asked to enter a number.
- The program prints the numbers from 0 to n-1.

```
n = int(input("Enter a number: "))
for i in range(n):
    print(i, end=" ")
```

break and continue

```
n = int(input("Enter a number: "))
for i in range(10):
    if i < n:
        print("The number is not found:", i)
        continue
    if i == n:
        print("The number is found:", i)
        break
```



```
r = random.randint(1, 10)
while True:
    n = int(input("Enter a number: "))
    if n == r:
        print("You guessed it right!")
        break
    elif n < r:
        print("Try a higher number.")
    else:
        print("Try a lower number.")
```

```
class range(stop)
class range(start, stop[, step])
```

The arguments to the range constructor must be integers (either built-in `int` or any object that implements the `__index__()` special method). If the `step` argument is omitted, it defaults to 1. If the `start` argument is omitted, it defaults to 0. If `step` is zero, `ValueError` is raised.

For a positive `step`, the contents of a range `r` are determined by the formula $r[i] = start + step * i$ where $i \geq 0$ and $r[i] < stop$.

For a negative `step`, the contents of the range are still determined by the formula $r[i] = start + step * i$, but the constraints are $i \geq 0$ and $r[i] > stop$.

start

The value of the `start` parameter (or 0 if the parameter was not supplied)

stop

The value of the `stop` parameter

step

The value of the `step` parameter (or 1 if the parameter was not supplied)

The advantage of the `range` type over a regular `list` or `tuple` is that a `range` object will always take the same (small) amount of memory, no matter the size of the range it represents (as it only stores the `start`, `stop` and `step` values, calculating individual items and subranges as needed).

Can we use while/for with else?

Week03/IntroductoryPythonDataStructures.pdf

INTRODUCTORY PYTHON : DATA STRUCTURES IN PYTHON

ASSOC. PROF. DR. BORA CANBULA
MANISA CELAL BAYAR UNIVERSITY

LISTS IN PYTHON:

Ordered and mutable sequence of values indexed by integers

Initializing

```
a_list = [] # empty
a_list = list() # empty
a_list = [3, 4, 5, 6, 7] # filled
```

Finding the index of an item

```
a_list.index(5) # 2 (the first occurrence)
```

Accessing the items

```
a_list[0] # 3
a_list[1] # 4
a_list[-1] # 7
a_list[-2] # 6
a_list[2:] # [5, 6, 7]
a_list[:2] # [3, 4]
a_list[1:4] # [4, 5, 6]
a_list[0:4:2] # [3, 5]
a_list[4:1:-1] # [7, 6, 5]
```

Adding a new item

```
a_list.append(9) # [3, 4, 5, 6, 7, 9]
a_list.insert(2, 8) # [3, 4, 8, 5, 6, 7, 9]
```

Update an item

```
a_list[2] = 1 # [3, 4, 1, 5, 6, 7, 9]
```

Remove the list or just an item

```
a_list.pop() # last item
a_list.pop(2) # with index
del a_list[2] # with index
a_list.remove(5) # first occurrence of 5
a_list.clear() # returns an empty list
del a_list # removes the list completely
```

Extend a list with another list

```
list_1 = [4, 2]
list_2 = [1, 3]
list_1.extend(list_2) # [4, 2, 1, 3]
```

Reversing and sorting

```
list_1.reverse() # [3, 1, 2, 4]
list_1.sort() # [1, 2, 3, 4]
```

Counting the items

```
list_1.count(4) # 1
list_1.count(5) # 0
```

Copying a list

```
list_1 = [3, 4, 5, 6, 7]
list_2 = list_1
list_3 = list_1.copy()
list_1.append(1)
list_2 # [3, 4, 5, 6, 7, 1]
list_3 # [3, 4, 5, 6, 7]
```

SETS IN PYTHON:

Unordered and mutable collection of values with no duplicate elements. They support mathematical operations like union, intersection, difference and symmetric difference

Initializing

```
a_set = set() # empty
a_set = {"name": "Bora"} # filled
```

No duplicate values

```
a_set.add(5) # (3, 4, 5)
```

Adding and updating the items

```
set_1 = {1, 3, 5}
set_2 = {5, 7, 9}
set_1.update(set_2) # {1, 3, 5, 7, 9}
```

Removing the items

```
a_set.pop() # removes an item and returns it
a_set.remove(3) # removes the item
a_set.discard(3) # removes the item
If item does not exist in set, remove() raises an error, discard() does not
a_set.clear() # returns an empty set
del a_set # removes the set completely
```

Mathematical operations

```
set_1 = {1, 2, 3, 5}
set_2 = {1, 2, 4, 6}
Union of two sets
set_1.union(set_2) # {1, 2, 3, 4, 5, 6}
```

Intersection of two sets

```
set_1.intersection(set_2) # {1, 2}
set_1 & set_2 # {1, 2}
```

Difference between two sets

```
set_1.difference(set_2) # {3, 5}
set_2.difference(set_1) # {4, 6}
```

Symmetric difference between two sets

```
set_1.symmetric_difference(set_2) # {3, 4, 5, 6}
```

Update sets with mathematical operations

```
set_1.intersection_update(set_2) # {1, 2}
set_1.difference_update(set_2) # {3, 5}
set_1.symmetric_difference_update(set_2) # {3, 4, 5, 6}
```

Copying a set

```
set_1 = {3, 4, 5, 6}
set_2 = set_1
```

Same as lists

DICTIONARIES IN PYTHON:

Unordered and mutable set of key-value pairs

Initializing

```
a_dict = {} # empty
a_dict = dict() # empty
a_dict = {"name": "Bora"} # filled
```

Accessing the items

```
a_dict["name"] # "Bora"
a_dict.get("name") # "Bora"
If the key does not exist in dictionary, index notation raises an error, get() method does not
```

Accessing the items with views

```
other_dict = {"a": 3, "b": 5, "c": 7}
other_dict.keys() # {'a', 'b', 'c'}
other_dict.values() # [3, 5, 7]
other_dict.items() # [('a', 3), ('b', 5), ('c', 7)]
```

Adding a new item

```
a_dict["city"] = "Manisa"
a_dict["age"] = 37
# {"name": "Bora", "city": "Manisa", "age": 37}
```

Update an item

```
a_dict["age"] = 38
# {"name": "Bora", "city": "Manisa", "age": 38}
other_dict = {"age": 39}
a_dict.update(other_dict)
# {"name": "Bora", "city": "Manisa", "age": 39}
```

Removing the items

```
a_dict.popitem() # last inserted item
a_dict.pop("city") # with a key
a_dict.clear() # returns an empty dictionary
del a_dict # removes the dict completely
```

Initialize a dictionary with fromkeys

```
a_list = ['a', 'b', 'c']
a_dict = dict.fromkeys(a_list)
# {'a': None, 'b': None, 'c': None}
a_dict = dict.fromkeys(a_list, 0)
# {'a': 0, 'b': 0, 'c': 0}
a_tuple = (3, 'name', 7)
a_dict = dict.fromkeys(a_tuple, True)
# {(3, 'name', 7): True, 'name': True, 7: True}
a_set = {0, 1, 2}
a_dict = dict.fromkeys(a_set, False)
# {(0, False, 1): False, 2: False}
```

Copying a set

```
set_1 = {3, 4, 5, 6}
```

TUPLES IN PYTHON:

Ordered and immutable sequence of values indexed by integers

Initializing

```
a_tuple = () # empty
a_tuple = tuple() # empty
a_tuple = (3, 4, 5, 6, 7) # filled
```

Finding the index of an item

```
a_tuple.index(5) # 2 (the first occurrence)
```

Accessing the items

```
Some index and slicing notation as lists
Adding, updating, and removing the items
Not allowed because tuples are immutable
```

Sorting

```
Tuples have no sort() method since they are immutable
sorted(a_tuple) # returns a sorted list
```

Counting the items

```
a_tuple.count(7) # 1
a_tuple.count(9) # 0
```

SOME ITERATION EXAMPLES:

```
a_list = [3, 5, 7]
a_tuple = (4, 6, 8)
a_set = {1, 4, 7}
a_dict = {"a": 1, "b": 2, "c": 3}
```

For ordered sequences

```
for i in range(len(a_list)):
    print(a_list[i])
for i, x in enumerate(a_tuple):
    print(i, x)
```

For ordered or unordered sequences

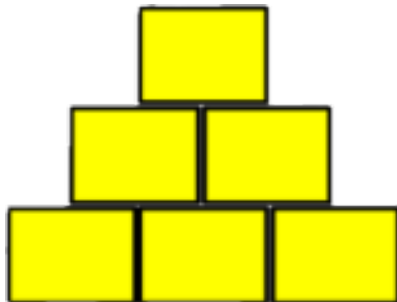
```
for a in a_set:
    print(a)
Only for dictionaries
for k in a_dict.keys():
    print(k)
for v in a_dict.values():
    print(v)
for k, v in zip(a_dict.keys(), a_dict.values()):
    print(k, v)
for k, v in a_dict.items():
    print(k, v)
```

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in-class quiz



Week03/pyramid_first_last.py



```
def calculate_pyramid_height(number_of_blocks):
```

```
    return height
```



Week03/sequences_first_last.py

```
def remove_duplicates(seq: list) -> list:
    """
    This function removes duplicates from a list.
    """
    return ...
```

```
def list_counts(seq: list) -> dict:
    """
    This function counts the number of
    occurrences of each item in a list.
    """
    return ...
```

```
def reverse_dict(d: dict) -> dict:
    """
    This function reverses the keys
    and values of a dictionary.
    """
    return ...
```

Exception Handling

Exception handling is a mechanism in Python to handle runtime errors gracefully without crashing the program.

```
try:
    # Code that may raise an exception
except Exception as e:
    # Code to handle the exception
else:
    # Code to execute if no exception is raised
finally:
    # Code that will be executed regardless of the result
```

Handling an Exception

```
try:
    1 / 0
except ZeroDivisionError:
    print("You cannot divide by zero.")
```

Handling All Exceptions

```
try:
    number = int(input("Enter a number: "))
    result = 1 / number
except Exception as e:
    print("An error occurred:", e)
```

Using finally Block

```
try:
    file = open("example.txt", "r")
    content = file.read()
except FileNotFoundError:
    print("File not found.")
else:
    print("File read successfully.")
    print("Content:", content)
finally:
    file.close()
    print("File closed.")
```

Handling Multiple Exceptions

```
try:
    number = int(input("Enter a number: "))
    result = 1 / number
except ValueError:
    print("Please enter a valid number.")
except ZeroDivisionError:
    print("You cannot divide by zero.")
```

Using else Block

```
try:
    number = int(input("Enter a number: "))
    result = 1 / number
except ValueError:
    print("Please enter a valid number.")
except ZeroDivisionError:
    print("You cannot divide by zero.")
else:
    print("The result is:", result)
```

Raising Exceptions

```
def calculate_area(radius):
    if radius < 0:
        raise ValueError("No negative radius!")
    return 3.14 * radius ** 2

try:
    area = calculate_area(-5)
except ValueError as e:
    print("An error occurred:", e)
```

Can we create custom exceptions?

Identifier Names

For variables, functions, classes etc. we use identifier names. We must obey some rules and we should follow some naming conventions.

Naming Conventions from PEP 8

- **Names to Avoid**

Never use the characters 'l' (lowercase letter el), 'O' (uppercase letter oh), or 'I' (uppercase letter eye) as single character variable names.

- **Packages**

Short, all-lowercase names without underscores

- **Modules**

Short, all-lowercase names, can have underscores

- **Classes**

CapWords (upper camel case) convention

- **Functions**

snake_case convention

- **Variables**

snake_case convention

- **Constants**

ALL_UPPERCASE, words separated by underscores

Underscore Usage in Identifier Names

- **`_single_leading_underscore`**

Weak “internal use” indicator.

from M import * does not import objects whose names start with an underscore.

- **`single_trailing_underscore_`**

Used by convention to avoid conflicts with keyword.

- **`__double_leading_underscore`**

When naming a class attribute, invokes name mangling (inside class FooBar, `__boo` becomes `_FooBar__boo`)

- **`__double_leading_and_trailing_underscore__`**

“magic” objects or attributes that live in user-controlled namespaces (`__init__`, `__import__`, etc.). Never invent such names; only use them as documented.

Functions

Functions are defined by using def keyword, name, and the parenthesized list of formats parameters.

Naming Convention from PEP8

Function names should be lowercase, with words separated by underscores as necessary to improve the readability.

Basic Function Definition

```
def function_name():  
    pass
```

Input and Output Arguments

```
def fn(arg1, arg2):  
    return arg1 + arg2
```

Default Values for Arguments

```
def fn(arg1 = 0, arg2 = 0):  
    return arg1 + arg2
```

Type Hints and Default Values for Arguments

```
def fn(arg1: int = 0, arg2: int = 0) -> int:  
    return arg1 + arg2
```

PEP 3107

Multiple Type Hints for Arguments (> Python 3.10)

```
def fn(arg1: int|float, arg2: int|float) -> (float, float):  
    return arg1 * arg2, arg1 / arg2
```

> Python 3.10

Lambda Functions

```
fn = lambda arg1, arg2: arg1 + arg2
```

Function Docstrings

```
def fn(arg1 = 0, arg2 = 0):  
    """This function sums two number."""  
    return arg1 + arg2
```

PEP 257

Docstrings **PEP 257**

A docstring is a string literal that occurs as the first statement in a module, function, class or method definition. Such a docstring becomes the `__doc__` special attribute of that object.

One-line Docstrings

```
def fn(arg1 = 0, arg2 = 0):  
    """This function sums two number."""  
    return arg1 + arg2
```

Multi-line Docstrings

```
def fn(arg1 = 0, arg2 = 0):  
    """This function sums two number.  
  
    Keyword arguments:  
    arg1 -- first number (default 0)  
    arg2 -- second number (default 0)  
    Return: the sum of arg1 and arg2  
    """"  
    return arg1 + arg2
```

Docutils and Sphinx are tools to automatically create documentations

reST (reStructuredText)

```
def fn(arg1 = 0, arg2 = 0):  
    """"  
    This function sums two number.  
  
    :param arg1: First number  
    :param arg2: Second number  
    :return: Sum of two numbers  
    """"  
    return arg1 + arg2
```

Google

```
def fn(arg1 = 0, arg2 = 0):  
    """"  
    This function sums two number.  
  
    Args:  
        arg1 (int): First number  
        arg2 (int): Second number  
  
    Returns:  
        int: Sum of two numbers  
    """"  
    return arg1 + arg2
```

Some other formats are Epytext (javadoc), Numpydoc, etc.

Parameter Kinds

PEP 362

Kind describes how argument values are bound to the parameter. The kind can be fixed in the signature of the function.

Positional-or-Keyword (Standard Binding)

```
def fn(arg1 = 0, arg2 = 0):  
    return arg1 + arg2
```

fn(), fn(3), fn(3, 5), fn(arg1=3), fn(arg2=5), fn(arg1=3, arg2=5)

Positional-or-Keyword and Keyword-Only

```
def fn(arg1 = 0, arg2 = 0, *, arg3 = 1):  
    return (arg1 + arg2) * arg3
```

fn(), fn(3), fn(3, 5), fn(3, 5, 2), fn(arg1=3), fn(arg2=5), fn(arg1=3, arg2=5),
fn(3, 5, arg3=2), fn(arg1=3, arg2=5, arg3=2), fn(arg3=2, arg1=3, arg2=5)

Positional-Only and Positional-or-Keyword and Keyword-Only

```
def fn(arg1=0, arg2=0, /, arg3=1, arg4=1, *, arg5=1, arg6=1):  
    return (arg1 + arg2) * arg3 / arg4 * arg5**arg6
```

fn(), fn(3), fn(3, 5), fn(3, 5, 2), fn(3, 5, 2, 4), fn(3, 5, 2, 4, 7),
fn(3, 5, arg3=2, arg4=4), fn(arg1=3, arg2=5, arg3=2, arg4=4),
fn(3, 5, arg3=2, arg4=4, arg5=7, arg6=8), fn(3, 5, 2, 4, arg5=7, arg6=8),
fn(arg1=3, arg2=5, arg3=2, arg4=4, arg5=7, arg6=8)

PEP 457

***args and **kwargs**

```
def fn(*args, **kwargs):  
    print(args) # a tuple of positional arguments  
    print(kwargs) # a dictionary of keyword arguments
```

fn(), fn(3), fn(3, 5), fn(x=3, y=5), fn(3, 5, x=2, y=4),
fn(3, 5, x=2, y=4, 1, 2)

Function Attributes

PEP 232

Functions already have a number of attributes such as `__doc__`, `__annotations__`, `__defaults__`, etc. Like everything in Python, functions are also objects, therefore, user can add a dictionary as attributes by using `get / set` methods to `__dict__`.

Built-in Functions

The Python interpreter has a number of functions and types built into it that are always available. They are listed here in alphabetical order.

Built-in Functions			
A <code>abs()</code> <code>aiter()</code> <code>all()</code> <code>anext()</code> <code>any()</code> <code>ascii()</code>	E <code>enumerate()</code> <code>eval()</code> <code>exec()</code>	L <code>len()</code> <code>list()</code> <code>locals()</code>	R <code>range()</code> <code>repr()</code> <code>reversed()</code> <code>round()</code>
B <code>bin()</code> <code>bool()</code> <code>breakpoint()</code> <code>bytearray()</code> <code>bytes()</code>	F <code>filter()</code> <code>float()</code> <code>format()</code> <code>frozenset()</code>	M <code>map()</code> <code>max()</code> <code>memoryview()</code> <code>min()</code>	S <code>set()</code> <code>setattr()</code> <code>slice()</code> <code>sorted()</code> <code>staticmethod()</code> <code>str()</code> <code>sum()</code> <code>super()</code>
C <code>callable()</code> <code>chr()</code> <code>classmethod()</code> <code>compile()</code> <code>complex()</code>	C <code>getattr()</code> <code>globals()</code>	N <code>next()</code>	T <code>tuple()</code> <code>type()</code>
D <code>delattr()</code> <code>dict()</code> <code>dir()</code> <code>divmod()</code>	H <code>hasattr()</code> <code>hash()</code> <code>help()</code> <code>hex()</code>	O <code>object()</code> <code>oct()</code> <code>open()</code> <code>ord()</code>	V <code>vars()</code>
	I <code>id()</code> <code>input()</code> <code>int()</code> <code>isinstance()</code> <code>issubclass()</code> <code>iter()</code>	P <code>pow()</code> <code>print()</code> <code>property()</code>	Z <code>zip()</code>
			<code>__import__()</code>

Function Attributes

PEP 232

Functions already have a number of attributes such as `__doc__`, `__annotations__`, `__defaults__`, etc. Like everything in Python, functions are also objects, therefore, user can add a dictionary as attributes by using `get / set` methods to `__dict__`.

setattr(*object*, *name*, *value*)

This is the counterpart of `getattr()`. The arguments are an object, a string, and an arbitrary value. The string may name an existing attribute or a new attribute. The function assigns the value to the attribute, provided the object allows it. For example, `setattr(x, 'foobar', 123)` is equivalent to `x.foobar = 123`.

name need not be a Python identifier as defined in [Identifiers and keywords](#) unless the object chooses to enforce that, for example in a custom `__getattr__()` or via `__slots__`. An attribute whose name is not an identifier will not be accessible using the dot notation, but is accessible through `getattr()` etc..

getattr(*object*, *name*)**getattr**(*object*, *name*, *default*)

Return the value of the named attribute of *object*. *name* must be a string. If the string is the name of one of the object's attributes, the result is the value of that attribute. For example, `getattr(x, 'foobar')` is equivalent to `x.foobar`. If the named attribute does not exist, *default* is returned if provided, otherwise `AttributeError` is raised. *name* need not be a Python identifier (see `setattr()`).

hasattr(*object*, *name*)

The arguments are an object and a string. The result is `True` if the string is the name of one of the object's attributes, `False` if not. (This is implemented by calling `getattr(object, name)` and seeing whether it raises an `AttributeError` or not.)

delattr(*object*, *name*)

This is a relative of `setattr()`. The arguments are an object and a string. The string must be the name of one of the object's attributes. The function deletes the named attribute, provided the object allows it. For example, `delattr(x, 'foobar')` is equivalent to `del x.foobar`. *name* need not be a Python identifier (see `setattr()`).

Nested Scopes

PEP 227

Function objects can have methods. These methods can be used as inner functions and can be useful for encapsulation.

```
def parent_function():  
    def nested_function():  
        print("Nested function")  
    print("Parent function")  
    parent_function.nested_function = nested_function
```



```
parent_function()  
parent_function.nested_function()
```

Getter and Setter Methods

```
def point(x, y):  
    def set_x(new_x):  
        nonlocal x  
        x = new_x  
    def set_y(new_y):  
        nonlocal y  
        y = new_y  
    def get():  
        return x, y  
    point.set_x = set_x  
    point.set_y = set_y  
    point.get = get  
    return point
```

Decorators

Decorators take a function as argument and returns a function. They are used to extend the behavior of the wrapper function, without modifying it. So they are very useful for dealing with code legacy.

Traditional Way

```
def my_decorator(fn):
    def _my_decorator():
        print("Before function")
        fn()
        print("After function")
    return _my_decorator

def my_decorated_function():
    print("Function")

my_decorated_function = \
    my_decorator(my_decorated_function)
```

Pythonic Way

```
def d1(fn):
    def _d1():
        print("Before d1")
        fn()
        print("After d1")
    return _d1

@d1
def f1():
    print("Function")
```

Decorators with Arguments

```
def decorator(func):
    def _decorator(*args, **kwargs):
        print("I am decorator")
        print(args)
        print(kwargs)
        func(*args, **kwargs)
    return _decorator

@decorator
def decorated_func_w_args(x):
    print(f"x = {x}")

@decorator
def decorated_triple_print(
    x=None, y=None, z=None):
    x_str = f"x = {x} " \
        if x is not None else f""
    y_str = f"y = {y} " \
        if y is not None else f""
    z_str = f"z = {z} " \
        if z is not None else f""
    print(x_str + y_str + z_str)
```

Decorator Chain

```
def d1(func):
    def _d1(*args, **kwargs):
        print(f"d1 here for \
            {func.__name__}")
        func(*args, **kwargs)
    return _d1

def d2(func):
    def _d2(*args, **kwargs):
        print(f"d2 here for \
            {func.__name__}")
        func(*args, **kwargs)
    return _d2

@d1
@d2
def fd(x):
    print(f"f says {x}")
```





Week04/functions_firstname_lastname.py

custom_power

- ✓ A lambda function
- ✓ Two parameters (x and e)
- ✓ x is positional-only
- ✓ e is positional-or-keyword
- ✓ x has the default value 0
- ✓ e has the default value 1
- ✓ Returns $x^{**}e$

custom_equation

- ✓ A function returns float
- ✓ Five integer parameters (x, y, a, b, c)
- ✓ x is positional-only with default value 0
- ✓ y is positional-only with default value 0
- ✓ a is positional-or-keyword with default value 1
- ✓ b is positional-or-keyword with default value 1
- ✓ c is keyword-only with default value 1
- ✓ Function signature must include all annotations
- ✓ Docstring must be in reST format.
- ✓ Returns $(x^{**}a + y^{**}b) / c$

fn_w_counter

- ✓ A function returns a tuple of an int and a dictionary
- ✓ Function must count the number of calls with caller information
- ✓ Returning integer is the total number of calls
- ✓ Returning dictionary with string keys and integer values includes the caller (`__name__`) as key, the number of call coming from this caller as value.

Examples

```
custom_power(2) == 2
custom_power(2, 3) == 8
custom_power(2, e=2) == 4
custom_equation(2, 3) == 5.0
custom_equation(2, 3, 2) == 7.0
custom_equation(2, 3, 2, 3) == 31.0
custom_equation(3, 5, a=2, b=3, c=4) == 33.5
custom_equation(3, 5, 2, b=3, c=4) == 33.5
custom_equation(3, 5, 2, 3, c=4) == 33.5
for i in range(10):
    fn_w_counter()
fn_w_counter() == (11, {'__main__': 11})
```



Week04/decorators_firstname_lastname.py

performance

- ✓ A decorator which measures the performance of functions and also saves some statistics.
- ✓ Has three attributes: counter, total_time, total_mem
- ✓ Attribute counter stores the number of times that the decorator has been called.
- ✓ Attribute total_time stores the number of total time that the functions took.
- ✓ Attribute total_mem stores the total memory in bytes that the functions consumed.

The problems in computer programming can be categorized based on the primary source of their performance bottlenecks.

I/O-bound Problems

While solving an **I/O-bound** problem, the system spends a significant amount of time waiting for input/output operations.

Subcategories can be **Disk I/O** (reading or writing to a hard drive) and **Network I/O** (waiting for data from a remote server).

The solutions often involve **asynchronous programming**, caching, or optimizing the I/O operations.

CPU-bound Problems

For **CPU-bound** problems, computational processing is the bottleneck.

Speeding up the computation requires either a faster CPU or optimizing the computation itself.

Parallel processing, **algorithm optimization**, or offloading computations to other systems or specialized hardware (like **GPUs**) are common strategies to overcome these problems.

Memory-bound Problems

Problems where the primary constraint is the **system's memory**.

Solutions can involve **optimizing data structures**, utilizing **external memory storage**, or employing algorithms that are more **memory-efficient**.

Coroutines are a generalization of subroutines (functions) used for **cooperative** multitasking. They can be called once and do not maintain any state between calls. They can be called on and later continue from where they left off.

Do you know any Callable which can pause its execution?

 **return**

Regular functions returning a specified value back to the caller and terminates the function's execution.

Once the function returns a value using **return**, **its state lost**. Subsequent calls to the function start the execution from the beginning of the function.

Used to compute a value and return it to caller immediately.

 **yield**

Used in special functions known as **generators**. Produces a series of values for iteration using a **lazy evaluation** approach (values are generated on-the-fly, not stored in memory).

When a function using the **yield** keyword is called, it returns a **generator** object without even beginning execution of the function.

Upon calling **next()**, the function runs until it encounters the **yield** keyword. The function's execution is **paused**, and the yielded value is returned. Subsequent calls to **next()** resume the function's execution immediately after the last **yield** statement.

Once all values have been yielded, the generator raises a **StopIteration** exception.

Can you write your own generator by using some magic methods in a class?



Coroutines are a generalization of subroutines (functions) used for **cooperative multitasking**. Unlike functions that return once and do not maintain state, **coroutines can pause** their execution and later continue from where they left off.

Traditional synchronous programming runs line by line.
In I/O-bound problems, the program waits for the operation.
Asynchronous programming allows tasks to run concurrently.

Increasing Efficiency

Can you increase the efficiency in real world problems with this technique?

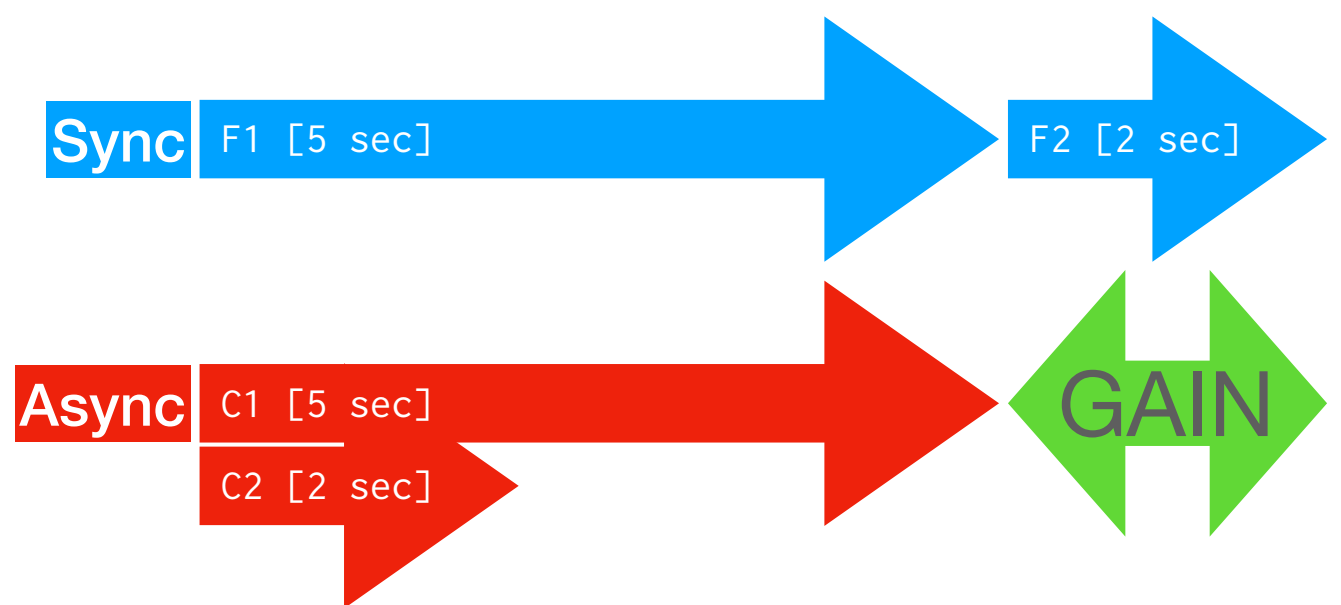


<https://youtu.be/loqCY9b7aec>
<https://www.canbula.com/cookie>

How to implement Coroutines in Python?

Coroutines declared with the `async/await` syntax is the best practice of writing asynchronous applications in Python.

- ✓ Handle many tasks concurrently without multi-threading.
- ✓ Improve performance for I/O-bound tasks.



Can you write your own awaitable by using some magic methods in a class?





<https://forms.cloud.microsoft/>

in-class quiz



Week05/awaitme_firstname_lastname.py

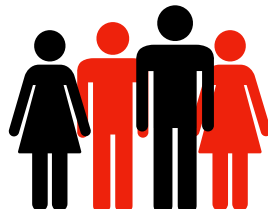
awaitme

- ✓ A decorator which turns any function into a coroutine.
- ✓ It must pass all the arguments to the function properly.
- ✓ If function returns any value, so the decorator returns it.



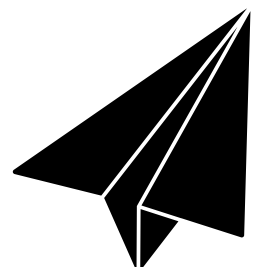
Rules for your pull requests

- ✓ Please run your code first in your computer, do not submit codes with syntax errors.
- ✓ Submit your code to WeekXX folder. WeekXX/hw is for me to move accepted works.
- ✓ If a change is requested, please edit the existing pull request, don't open a new one.



PROJECT: Implement the cookie problem in Python

- ✓ Application optimizes any recipe.
- ✓ You can group up to 4 students.
- ✓ Backend with Python.
- ✓ Tests with Pytest.
- ✓ Frontend with HTML + CSS + JS.
- ✓ 5 minutes presentation in English.



Wrap-up: Use what we have learned so far in a project!

Using requests module

```
Python 3.12.7 (main, Oct 1 2024, 02:05:46) [Clang 16.0.0 (clang-1600.0.26.3)] on darwin
Type "help", "copyright", "credits" or "license" for more information.
>>> import requests
>>> dir(requests)
['ConnectTimeout', 'ConnectionError', 'DependencyWarning', 'FileModeWarning', 'HTTPError',
 'JSONDecodeError', 'NullHandler', 'PreparedRequest', 'ReadTimeout', 'Request', 'RequestEx
 ception', 'RequestsDependencyWarning', 'Response', 'Session', 'Timeout', 'TooManyRedirects
 ', 'URLRequired', '__author__', '__author_email__', '__build__', '__builtins__', '__cached
 __', '__cake__', '__copyright__', '__description__', '__doc__', '__file__', '__license__',
 '__loader__', '__name__', '__package__', '__path__', '__spec__', '__title__', '__url__',
 '__version__', '_check_cryptography', '_internal_utils', 'adapters', 'api', 'auth', 'certs
 ', 'chardet_version', 'charset_normalizer version', 'check_compatibility', 'codes', 'compa
 t', 'cookies', 'delete', 'exceptions', 'get', 'head', 'hooks', 'logging', 'models', 'optio
 ns', 'packages', 'patch', 'post', 'put', 'request', 'session', 'sessions', 'ssl', 'status_
 codes', 'structures', 'urllib3', 'utils', 'warnings']
>>> help(requests.get)
```

get(url, params=None, **kwargs)
Sends a GET request.

:param url: URL for the new
:param params: (optional) D
in the query string for
:param **kwargs: Optional
:return: :class:`Response`
:rtype: **requests.Response**

NAME	requests.exceptions
DESCRIPTION	requests.exceptions ~~~~~ This module contains the set of Requests' exceptions.
CLASSES	builtins.OSError(builtins.Exception) RequestException ChunkedEncodingError ConnectionError ConnectTimeout(ConnectionError, Timeout) ProxyError SSLError ContentDecodingError(RequestException, urllib3.exceptions.HTTPError) HTTPError InvalidHeader(RequestException, builtins.ValueError) InvalidJSONError JSONDecodeError(InvalidJSONError, json.decoder.JSONDecodeError) InvalidSchema(RequestException, builtins.ValueError) InvalidURL(RequestException, builtins.ValueError) InvalidProxyURL MissingSchema(RequestException, builtins.ValueError) RetryError StreamConsumedError(RequestException, builtins.TypeError) Timeout ReadTimeout TooManyRedirects URLRequired UnrewindableBodyError builtins.Warning(builtins.Exception) RequestsWarning FileModeWarning(RequestsWarning, builtins.DeprecationWarning) RequestsDependencyWarning

to send

```
>>> r = requests.get("https://www.cant")
dir(r)
['_attrs_', '_bool_', '_class_',
 '_enter_', '_eq_', '_exit_', '_f
_', '_gt_', '_hash_', '_init_',
'_module_', '_ne_', '_new_',
'_setattr_', '_setstate_',
_, '_content', '_content_consumed', '_next', 'apparent_encoding', 'close', 'connection',
'content', 'cookies', 'elapsed', 'encoding', 'headers', 'history', 'is_permanent_redirect',
'is_redirect', 'iter_content', 'iter_lines', 'json', 'links', 'next', 'ok', 'raise_for_s
tatus', 'raw', 'reason', 'request', 'status_code', 'text', 'url']
>>> help(r.json)
```

doc_', '
getstate_
'_lt_',
'_repr_
weakref_

json(kwargs)** method of requests.models.Response instance
Returns the json-encoded content of a response, if any.

:param **kwargs: Optional arguments that ``json.loads`` takes.
:raises requests.exceptions **JSONDecodeError**: If the response body does not contain valid json.

Wrap-up: Use what we have learned so far in a project!

Using logging module

```
Python 3.12.7 (main, Oct 1 2024, 02:05:46) [Clang 16.0.0 (clang-1600.0.26.3)] on darwin
Type "help", "copyright", "credits" or "license" for more information.
>>> import logging
>>> print([s for s in dir(logging) if not s.startswith("_")])
['BASIC_FORMAT', 'BufferingFormatter', 'CRITICAL', 'DEBUG', 'ERROR', 'FATAL', 'FileHandler', 'Filter', 'Filterer', 'Formatter', 'GenericAlias', 'Handler', 'INFO', 'LogRecord', 'Logger', 'LoggerAdapter', 'Manager', 'NOTSET', 'NullHandler', 'PercentStyle', 'Placeholder', 'RootLogger', 'StrFormatStyle', 'StreamHandler', 'StringTemplateStyle', 'Template', 'WARN', 'WARNING', 'addLevelName', 'atexit', 'basicConfig', 'captureWarnings', 'collections', 'critical', 'currentframe', 'debug', 'disable', 'error', 'exception', 'fatal', 'getHandlerByName', 'getHandlerNames', 'getLevelName', 'getLevelNamesMapping', 'getLogRecordFactory', 'getLogger', 'getLoggerClass', 'info', 'io', 'lastResort', 'log', 'logAsyncioTasks', 'logMultiprocessing', 'logProcesses', 'logThreads', 'makeLogRecord', 'os', 'raiseExceptions', 're', 'root', 'setLogRecordFactory', 'setLoggerClass', 'shutdown', 'sys', 'threading', 'time', 'traceback', 'warn', 'warning', 'warnings', 'weakref']
```

```
debug(msg, *args, **kwargs)
Log a message with severity 'DEBUG' on the root logger. If the logger has no handlers, call basicConfig() to add a console handler with a pre-defined format.
```

```
basicConfig(**kwargs)
Do basic configuration for the logging system.

This function does nothing if the root logger already has handlers configured, unless the keyword argument *force* is set to `True`. It is a convenience method intended for use by simple scripts to do one-shot configuration of the logging package.

The default behaviour is to create a StreamHandler which writes to sys.stderr, set a formatter using the BASIC_FORMAT format string, and add the handler to the root logger.

A number of optional keyword arguments may be specified, which can alter the default behaviour.

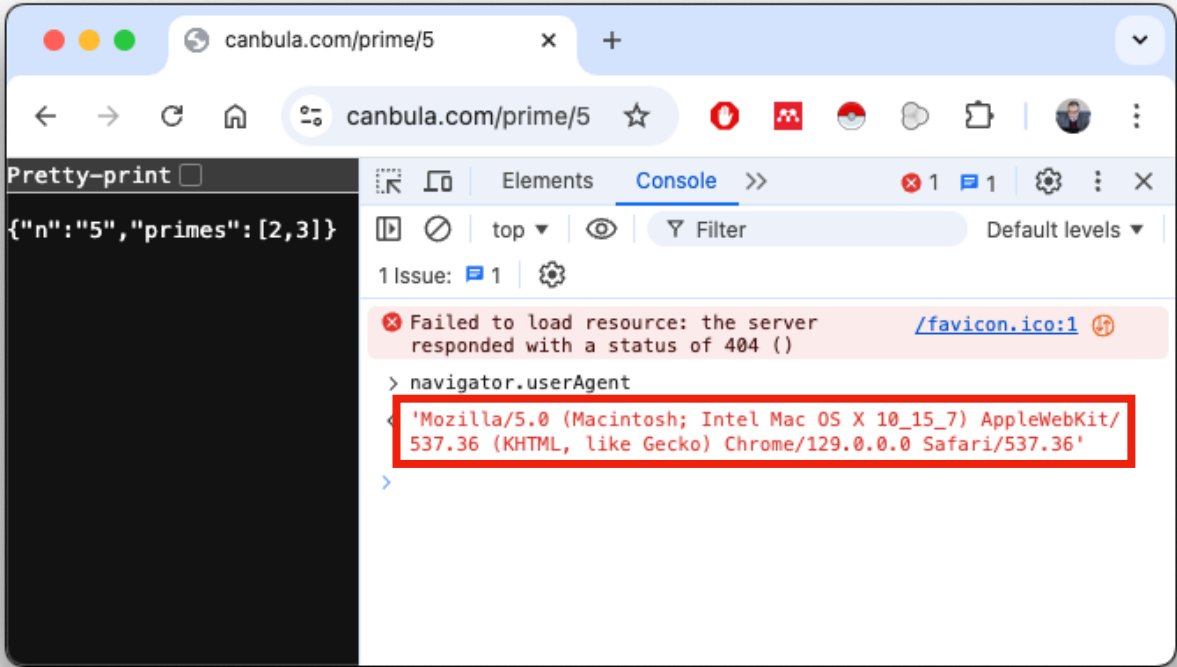
filename Specifies that a FileHandler be created, using the specified filename, rather than a StreamHandler.
filemode Specifies the mode to open the file, if filename is specified (if filemode is unspecified, it defaults to 'a').
format Use the specified format string for the handler
datefmt Use the specified date/time format.
style If a format string is specified, use the type of format string (possible values: %-formatting, :meth:`str.format` and :code:`%` - defaults to `%`).
level Set the root logger level to the specified level.
stream Use the specified stream to initialize the root logger. If this argument is incompatible with any handlers already present, 'stream' is ignored.
handlers If specified, this should be an iterable of handlers, which will be added to the root logger in the list which does not have a formatter assigned the formatter created in this function.
force If this keyword is specified as true, any existing handlers attached to the root logger are removed and closed, before carrying out the configuration as specified by the other arguments.
encoding If specified together with a filename, this encoding is passed to the created FileHandler, causing it to be used when the file is opened.
errors If specified together with a filename, this value is passed to the created FileHandler, causing it to be used when the file is opened in text mode. If not specified, the default value is 'backslashreplace'.
```

Logging Levels		
The numeric values of logging levels are given in the following table. These are primarily of interest if you want to define your own levels, and need them to have specific values relative to the predefined levels. If you define a level with the same numeric value, it overwrites the predefined value; the predefined name is lost.		
Level	Numeric value	What it means / When to use it
logging.NOTSET	0	When set on a logger, indicates that ancestor loggers are to be consulted to determine the effective level. If that still resolves to NOTSET, then all events are logged. When set on a handler, all events are handled.
logging.DEBUG	10	Detailed information, typically only of interest to a developer trying to diagnose a problem.
logging.INFO	20	Confirmation that things are working as expected.
logging.WARNING	30	An indication that something unexpected happened, or that a problem might occur in the near future (e.g. 'disk space low'). The software is still working as expected.
logging.ERROR	40	Due to a more serious problem, the software has not been able to perform some function.
logging.CRITICAL	50	A serious error, indicating that the program itself may be unable to continue running.

```
logging.basicConfig(
    format="%(levelname)s @ %(asctime)s : %(message)s",
    datefmt="%d.%m.%Y %H:%M:%S",
    level=logging.INFO,
    handlers=[
        logging.FileHandler("requests.log", mode="w"),
        logging.StreamHandler()
    ],
)
```

Wrap-up: Use what we have learned so far in a project!

Solving the problems of requests module



```
class Session(SessionRedirectMixin)
    """A Requests session.

    Provides cookie persistence, connection-pooling,
    and more.

    Basic Usage::

    >>> import requests
    >>> s = requests.Session()
    >>> s.get('https://httpbin.org/get')
    <Response [200]>

    Or as a context manager::

    >>> with requests.Session() as s:
    ...     s.get('https://httpbin.org/get')
    <Response [200]>

    Method resolution order:
    Session
    SessionRedirectMixin
    builtins.object

    Methods defined here:
    __enter__(self)
    __exit__(self, *args)
    __getstate__(self)
        Helper for pickle.
    __init__(self)
        Initialize self. See help(type(self)) for details.
    __setstate__(self, state)
```

```
request(self, method, url, params=None, data=None, headers=None, cookies=None,
        timeout=None, allow_redirects=True, proxies=None, hooks=None, stream=None,
        json=None)
    """Constructs a :class:`Request` object, prepares it and sends it.
    Returns :class:`Response` object.

    :param method: method for the new :class:`Request` object.
    :param url: URL for the new :class:`Request` object.
    :param params: (optional) Dictionary or bytes to be sent in the query
        string for the :class:`Request`.
    :param data: (optional) Dictionary, list of tuples, bytes, or file-like
        object to send in the body of the :class:`Request`.
    :param json: (optional) json to send in the body of the :class:`Request`.
    :param headers: (optional) Dictionary of HTTP Headers to send with the
        :class:`Request`.
    :param cookies: (optional) Dict or CookieJar object to send with the
        :class:`Request`.
    :param files: (optional) Dictionary of ``('filename': file-like-object)``
        for multipart encoding upload.
    :param auth: (optional) Auth tuple or callable to enable
        Basic/Digest/Custom HTTP Auth.
    :param timeout: (optional) How long to wait for the server to send
        data before giving up, as a float, or a :ref:`(connect timeout,
        read timeout)` tuple.
    :type timeout: float or tuple
    :param allow_redirects: (optional) Set to True by default.
    :type allow_redirects: bool
    :param proxies: (optional) Dictionary mapping protocol or protocol and
        hostname to the URL of the proxy.
    :param stream: (optional) whether to immediately download the response
        content. Defaults to ``False``.
    :param verify: (optional) Either a boolean, in which case it controls whether
        the server's TLS certificate is verified, or a string, in which case it
        must be a path to a CA bundle to use. Defaults to ``True``. When set to
        ``False``, requests will accept any TLS certificate presented by
        the server and will ignore hostname mismatches and/or expired
        certificates, which will make your application vulnerable to
        man-in-the-middle (MitM) attacks. Setting verify to ``False``
        may be useful during local development or testing.
    :param cert: (optional) If String, path to ssl client cert file (.pem).
        If Tuple, ('cert', 'key') pair.
    :rtype: requests.Response
```

Context Manager



<https://forms.cloud.microsoft/>

in-class quiz



Week06/timer_firstname_lastname.py

Timer

- ✓ Create a class Timer that measures the time taken by the block of code it manages.
- ✓ Timer class must be a context manager.
- ✓ The class must have two public attributes start_time and end_time, which are for the starting and the ending times, respectively.



Rules for your pull requests

- ✓ Please run your code first in your computer, do not submit codes with syntax errors.
- ✓ Submit your code to WeekXX folder. WeekXX/hw is for me to move accepted works.
- ✓ If a change is requested, please edit the existing pull request, don't open a new one.



**Good Luck
for your midterm!**

